Notes 2/5

Informal Exercise 32)

if $\emptyset$ is in our iterative definition

$3^2 = 9 = M_3^2(0)\ M_3 \circ M_3(\emptyset) = M_3(3 \cdot 0) = 3 \cdot 3 \cdot 0 = \emptyset$

correct but by definition $\emptyset$

Correct Formula $M_3^2(1)\ M_3 \circ M_3(1) = 3 \cdot 3 \cdot 1 = 9$ ✓

33) Commutative means $a * b = b * a$

$3^2 = 2^3$? NOPE $\neq$ commutative

Associative means $(a * b) * c = a * (b * c)$

$a = 2$
$b = 3$
$c = 2$

$(2^3)^2 = (2)^{(3^2)}$

$= 8^2 = 2^9$ NOPE $\neq$ ASSOCIATIVE

$= 64 \neq 512$

GOING INTO CHAPTER 2 NOW...

You can assume everything you have done in Chapter 1 to prove in chapter 2.

Site (By chapter 1).

Homework 2 Chapter 2

Claim. If $(S,<)$ satisfies 1,2,3 of Lemma 1 then it is a strict linear order (Satisfies Def 1)

1 of lemma => 1 of definition

Let $x,y \in S$ suppose at least one of $x<y$, $x=y$, $x>y$ holds. We are worried 2 might hold true.

Case 1: Suppose $x<y$ and $x=y$. This implies $x<y=x$. Thus $x<x$ which violate the anti reflexive law

Chapter 2: The Natural #'s as an ordered set

Def. 2 Associative to a strict linear order $<$ on a set S we define a nonstrict order relation $\leq$ by defining for all $x,y \in S$

$$x \leq y \overset{def}{\Longleftrightarrow} x<y \text{ or } x=y$$

Thm 2: If $x<y$ & $y \leq z$, then $x<z$ mixed transitivity

Also if $x<y$ & $y<z$ then $x<z$

Proof: Suppose $x<y$ & $y \leq z$. We have 2 cases $y=z$ & $y<z$

if $y<z$, then our normal transitivity shows $x<z$

Conversely, if $y=z$, then $x<y=z$, so $x<z$ ✓